

CSE207

Data Structure Practice Problems-1

Summer 2026

Course	CSE207	Topic	Array Operations
Language	C or C++	Type	In-Lab Practice

Instructions: Solve each problem using arrays. Take input from the user, show meaningful output, and handle valid array positions carefully.

Question 1: Insert an Element in an Array

Objective: Practice shifting elements to make space at a specific position.

Task: Write a program to insert a new element at a given 1-based position in an array.

Example	Array: 10 20 30 40 50; insert 25 at position 3 -> 10 20 25 30 40 50
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Question 2: Delete an Element from an Array

Objective: Practice shifting elements left after removing one item.

Task: Write a program to delete the element from a given 1-based position in an array.

Example	Array: 5 10 15 20 25; delete position 4 -> 5 10 15 25
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Question 3: Find the Largest and Smallest Element

Objective: Practice traversing an array and comparing values.

Task: Write a program to find and print both the largest and smallest elements in an array.

Example	Array: 12 45 2 67 34 -> largest = 67, smallest = 2
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Question 4: Reverse an Array In-Place

Objective: Practice swapping elements without using another array.

Task: Write a program to reverse the array by swapping the first and last elements, then moving inward.

Example	Array: 1 2 3 4 5 -> 5 4 3 2 1
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Question 5: Linear Search in an Array

Objective: Practice searching sequentially and reporting the correct position.

Task: Write a program to search for a given value. If found, print its 1-based position; otherwise print Not found.

Example	Array: 8 16 24 32 40; search 24 -> Element found at position 3
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Question 6: Rotate an Array by K Positions

Objective: Practice repeated shifts or a more efficient reversal-based method.

Task: Write a program to rotate an array to the right by K positions. If K is greater than the array size, use $K \% n$.

Example	Array: 1 2 3 4 5 6; K = 2 -> 5 6 1 2 3 4
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Question 7: Remove Duplicate Elements

Objective: Practice nested loops, element comparison, and compacting an array.

Task: Write a program to remove duplicate values from an unsorted array and print the array containing only the first occurrence of each value.

Example	Array: 4 2 4 7 2 9 7 -> 4 2 7 9
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